

# Car Seat Sales Forecasting Using Neural Networks

Pranai Basava

January 07, 2026

## Contents

|                                                        |   |
|--------------------------------------------------------|---|
| Executive Summary . . . . .                            | 1 |
| 1. Project Overview . . . . .                          | 1 |
| 2. Business Objective . . . . .                        | 2 |
| 3. Dataset Overview . . . . .                          | 2 |
| Target Variable . . . . .                              | 2 |
| Predictor Variables . . . . .                          | 2 |
| 4. Environment Setup . . . . .                         | 2 |
| 5. Data Ingestion . . . . .                            | 2 |
| 6. Data Preparation . . . . .                          | 2 |
| 7. Modeling Strategy . . . . .                         | 3 |
| Why Neural Networks? . . . . .                         | 3 |
| Validation Approach . . . . .                          | 3 |
| 8. Hyperparameter Tuning Strategy . . . . .            | 3 |
| 9. Model Training – Neural Network . . . . .           | 4 |
| 10. Hyperparameter Interpretation . . . . .            | 5 |
| 11. Generating Predictions on New Store Data . . . . . | 5 |
| 12. Business Insights . . . . .                        | 6 |
| 13. Limitations and Next Steps . . . . .               | 6 |

## Executive Summary

This analysis applies a **Neural Network regression model** to forecast car seat sales using store-level demographic, pricing, and merchandising attributes.

Neural Networks extend tree-based models by learning **complex non-linear relationships** through weighted combinations of input features. Using **5-fold cross-validation**, the model is trained and tuned to balance predictive accuracy and generalization performance.

The trained model is applied to **new, unseen store data** to generate forward-looking sales forecasts that can support inventory planning, pricing strategy, and demand estimation.

This model complements CART and Random Forest approaches by prioritizing **model flexibility and expressive power**, making it well-suited for capturing subtle patterns in retail sales data.

---

## 1. Project Overview

Accurate demand forecasting is essential for optimizing retail operations, including inventory management, pricing decisions, and marketing investments.

This notebook implements a **Neural Network regression model** to forecast car seat sales. The analysis follows the same dataset, validation framework, and evaluation strategy used in other models within this

project to ensure consistent and fair comparison.

---

## 2. Business Objective

The objectives of this analysis are to:

- Forecast car seat sales across retail locations
  - Capture complex, non-linear relationships in the data
  - Improve predictive performance beyond tree-based models
  - Generate reliable sales estimates for new store configurations
- 

## 3. Dataset Overview

The dataset contains store-level attributes related to competition, demographics, pricing, and merchandising.

### Target Variable

- Sales – Number of car seats sold (measured in hundreds of units)

### Predictor Variables

- **CompPrice** – Competitor pricing
- **Income** – Local average income (in \$000s)
- **Advertising** – Advertising spend (in \$000s)
- **Population** – Local population size (in \$000s)
- **Price** – Store price
- **ShelfLoc** – Shelf location quality
- **Age** – Average customer age
- **Education** – Average education level
- **Urban** – Urban location indicator
- **US** – U.S. store indicator

The **Store** variable is excluded from modeling as it serves only as a unique identifier.

---

## 4. Environment Setup

```
library(caret)
library(dplyr)
```

## 5. Data Ingestion

```
car_seat_sales <- read.csv("data/car_seat_sales.csv")
new_car_seat_sales <- read.csv("data/new_car_seat_sales_data.csv")
```

---

## 6. Data Preparation

The **Store** variable is removed prior to modeling, as it serves only as a unique identifier.

Neural Networks require numeric input; factor variables are automatically handled by **caret** through internal encoding.

```
model_data <- car_seat_sales %>%
  select(-Store)

new_model_data <- new_car_seat_sales %>%
  select(-Store)
```

---

## 7. Modeling Strategy

### Why Neural Networks?

Neural Networks are capable of learning highly non-linear patterns through weighted connections between input features and hidden layers.

Key advantages include:

- Flexibility in modeling complex relationships
- Ability to capture subtle feature interactions
- Strong performance in non-linear forecasting tasks

### Validation Approach

A 5-fold cross-validation strategy is employed to:

- Reduce overfitting risk
  - Stabilize performance estimates
  - Enable fair comparison across models
- 

## 8. Hyperparameter Tuning Strategy

Neural Networks require tuning of two key hyperparameters:

- **size** – Number of neurons in the hidden layer
- **decay** – Regularization parameter to prevent overfitting

```
set.seed(1975)

cv_method <- trainControl(
  method = "cv",
  number = 5
)

nn_grid <- expand.grid(
  .size = seq(5, 20, 5),
  .decay = seq(0.1, 0.5, 0.1)
)

nn_grid

##      .size .decay
```

```
## 1      5      0.1
## 2     10     0.1
## 3     15     0.1
## 4     20     0.1
## 5      5     0.2
## 6     10     0.2
## 7     15     0.2
## 8     20     0.2
## 9      5     0.3
## 10    10     0.3
## 11    15     0.3
## 12    20     0.3
## 13     5     0.4
## 14    10     0.4
## 15    15     0.4
## 16    20     0.4
## 17     5     0.5
## 18    10     0.5
## 19    15     0.5
## 20    20     0.5
```

---

## 9. Model Training – Neural Network

```
nn_model <- train(
  Sales ~ .,
  data = model_data,
  method = "nnet",
  trControl = cv_method,
  tuneGrid = nn_grid,
  linout = TRUE,
  trace = FALSE
)

nn_model

## Neural Network
##
## 400 samples
## 10 predictor
##
## No pre-processing
## Resampling: Cross-Validated (5 fold)
## Summary of sample sizes: 321, 319, 321, 320, 319
## Resampling results across tuning parameters:
##
##   size  decay  RMSE      Rsquared  MAE
##   5     0.1    5.263145  0.027933555  4.465051
##   5     0.2    5.214267  0.005100979  4.456274
##   5     0.3    5.324153  0.017165456  4.546040
##   5     0.4    5.340652  0.012123837  4.544519
##   5     0.5    5.283673  0.001733318  4.470238
##  10    0.1    5.349676  0.002825405  4.569844
```

```
## 10 0.2 5.293279 0.023905321 4.474850
## 10 0.3 5.232005 0.007269544 4.470904
## 10 0.4 5.197875 0.008128550 4.455160
## 10 0.5 5.251002 0.006025414 4.461578
## 15 0.1 5.238139 0.008339774 4.491886
## 15 0.2 5.295203 0.006676247 4.469122
## 15 0.3 5.479176 0.009587888 4.608825
## 15 0.4 5.308041 0.005488444 4.544203
## 15 0.5 5.377702 0.006539783 4.565746
## 20 0.1 5.336141 0.012045934 4.531243
## 20 0.2 5.415365 0.012371888 4.621312
## 20 0.3 5.268884 0.008909335 4.498215
## 20 0.4 5.324019 0.007982948 4.507340
## 20 0.5 5.467756 0.013131281 4.622393
##
## RMSE was used to select the optimal model using the smallest value.
## The final values used for the model were size = 10 and decay = 0.4.
```

---

## 10. Hyperparameter Interpretation

- Hidden layer size controls model capacity
- Decay penalizes large weights to reduce overfitting

The optimal combination of `size` and `decay` is selected based on minimum cross-validated RMSE, ensuring the model generalizes well to unseen data.

```
nn_model$bestTune
```

```
## size decay
## 9 10 0.4
```

---

## 11. Generating Predictions on New Store Data

In a production scenario, the trained Neural Network model is applied to new store-level data without historical sales information.

```
nn_predictions <- predict(nn_model, new_model_data)
```

```
prediction_results <- new_car_seat_sales %>%
  mutate(Predicted_Sales = nn_predictions)
```

```
head(prediction_results)
```

```
## Store CompPrice Income Advertising Population Price ShelfLoc Age Education
## 1 216 116 83 15 170 144 Bad 71 11
## 2 8 136 81 15 425 120 Good 67 10
## 3 273 113 33 0 14 63 Good 38 12
## Urban US Predicted_Sales
## 1 Yes Yes 9.9771709
## 2 Yes Yes 8.1118470
## 3 Yes No 0.9616064
```

`Predicted_Sales` represents the expected number of car seats sold, measured in **hundreds of units**, for each new store configuration.

---

## 12. Business Insights

Neural Networks provide a flexible modeling framework capable of capturing complex, non-linear demand patterns.

From a business perspective, this enables:

- Improved forecasting in dynamic retail environments
- Strong performance where linear or tree-based models may underfit
- Enhanced sensitivity to subtle pricing and demographic effects

However, Neural Networks trade interpretability for predictive power, making them most suitable when **accuracy is prioritized over transparency**.

---

## 13. Limitations and Next Steps

While Neural Networks offer strong predictive capability, they:

- Require careful hyperparameter tuning
- Are less interpretable than tree-based models
- May be sensitive to feature scaling and data distribution

Future work could include feature engineering, alternative network architectures, and ensemble strategies combining multiple model types.